

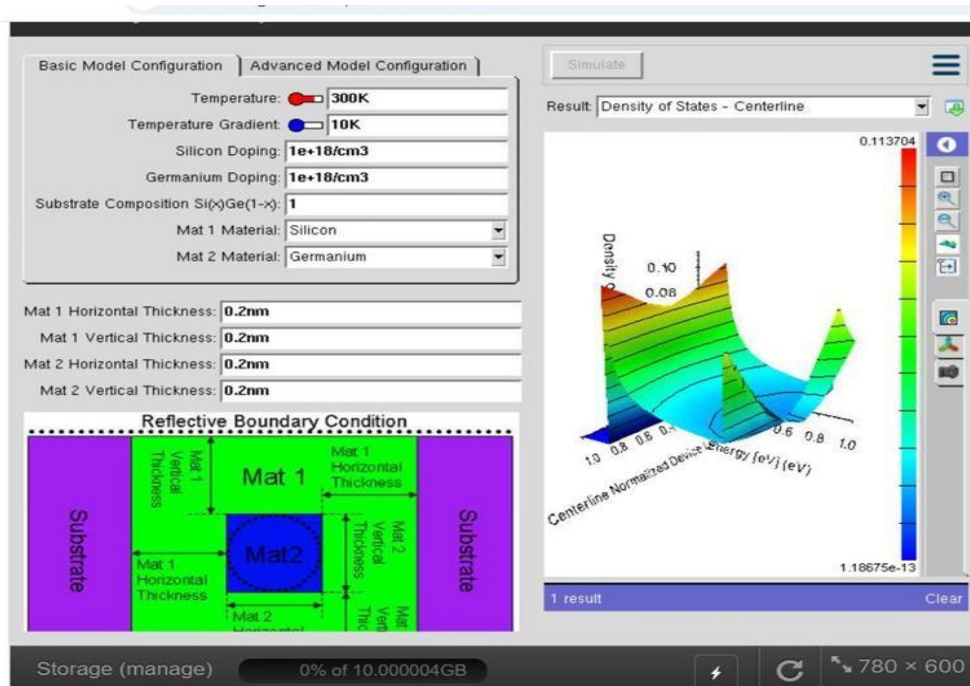
SLOT C-NANO ELECTRONICS

EXPERIMENT 2 - SIMULATION OF DENSITY OF STATES OF QUANTUM WELL STRUCTURE WITH VARIOUS NANOMETER THICKNESS

Aim: To simulate density of states of quantum well structure with various nano meter thickness.

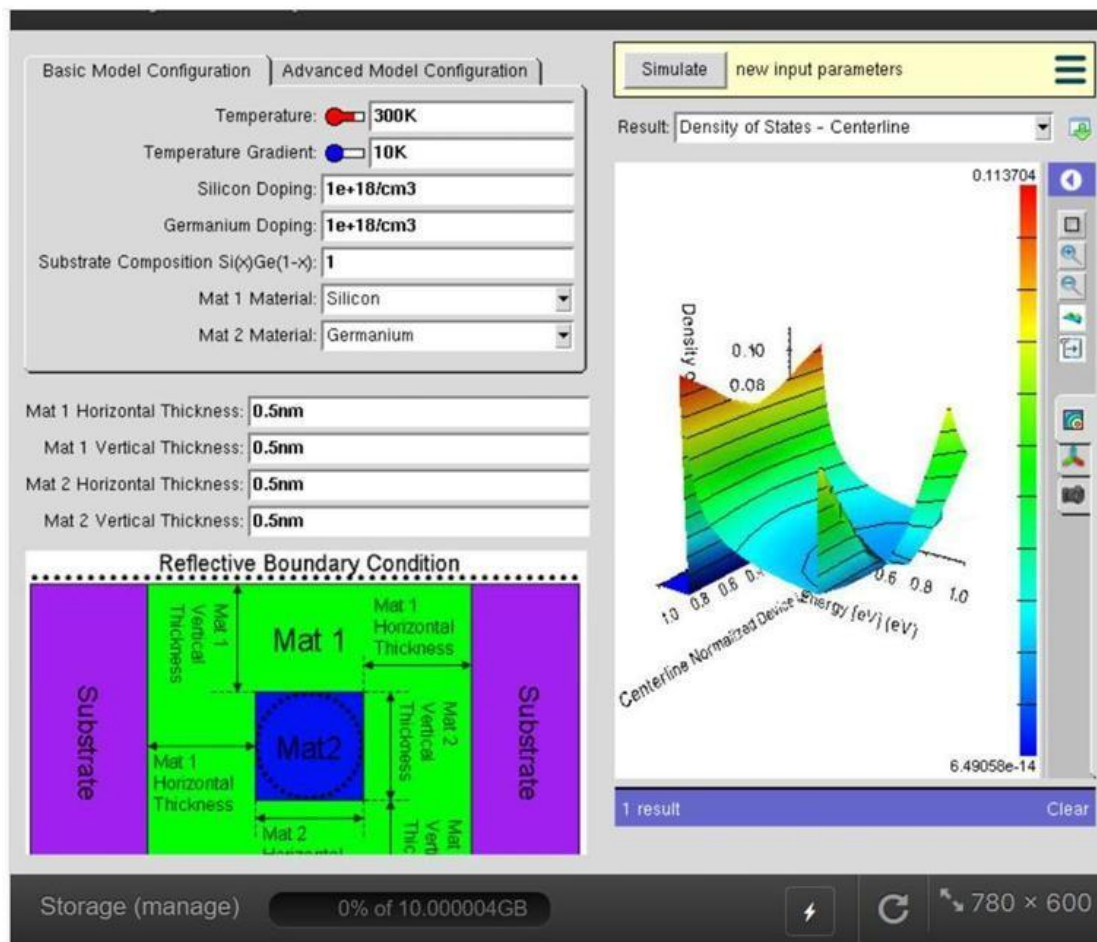
A quantum well is a potential well with only discrete energy values. The classic model used to demonstrate a quantum well is to confine particles, which were originally free to move in three dimensions, to two dimensions, by forcing them to occupy a planar region. The effects of quantum confinement take place when the quantum well thickness becomes comparable to the de Broglie wavelength of the carriers (generally electrons and holes), leading to energy levels called "energy subbands", i.e., the carriers can only have discrete energy values.

OUTPUT : Simulation of the density of states of quantum well of thickness 0.2 nm



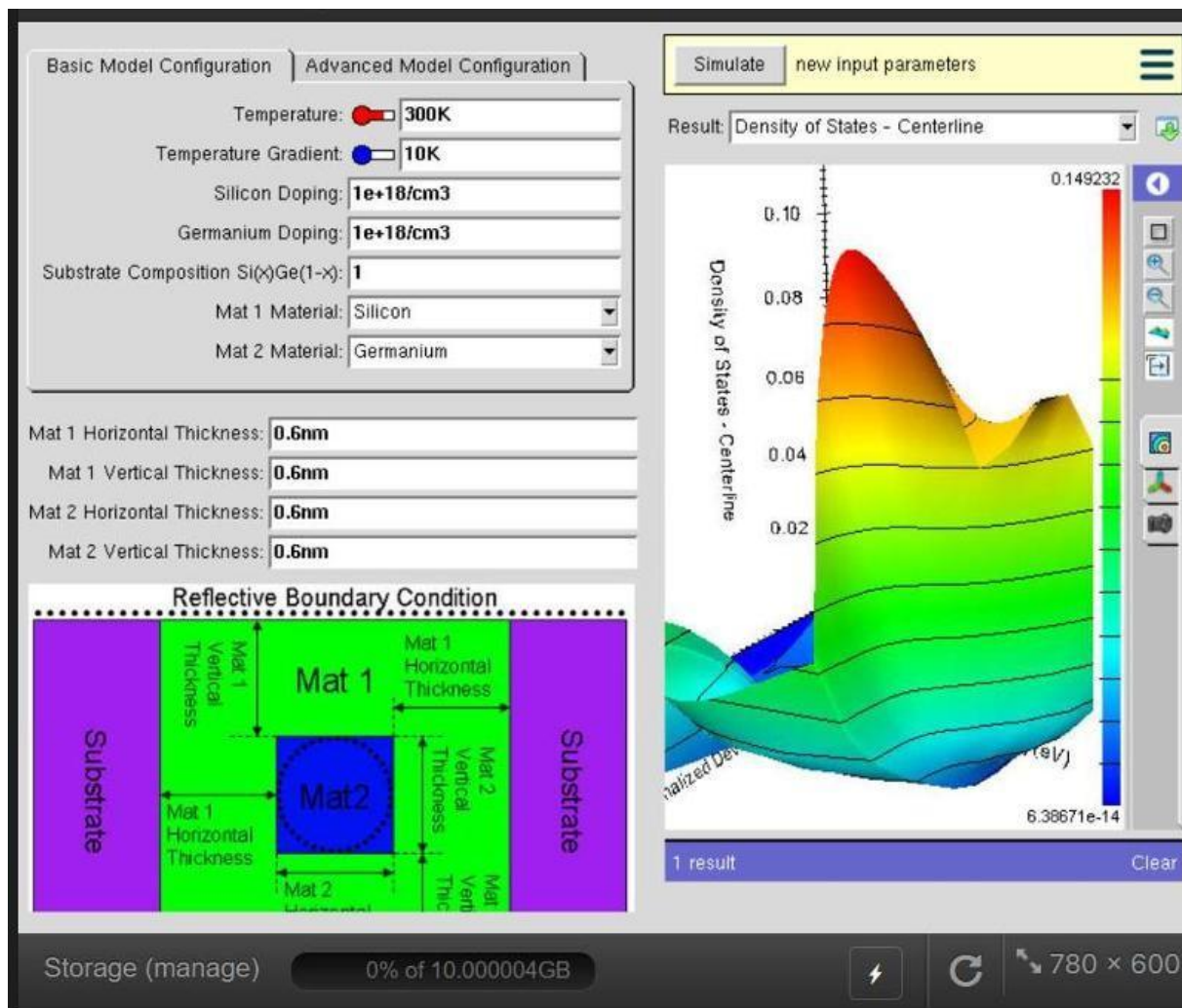
The output shows the density of states of a 0.2 nm quantum well. Strong quantum confinement creates discrete energy levels, producing a step-like DOS. As energy increases, more sub-bands contribute, increasing the density of states. This highlights the effect of ultra-thin quantum wells on electronic behavior.

CASE STUDY 1: simulate the density of states of quantum well of thickness 0.5nm



The simulation illustrates the density of states of a **0.5 nm quantum well**. Due to quantum confinement, the energy levels are quantized, producing a **step-like DOS**. The increased well thickness reduces confinement compared to thinner wells, resulting in more closely spaced energy levels.

CAE STUDY 2: simulate the density of states of quantum well of thickness 0.6 nm



Conclusion:

The simulation shows that a 0.5 nm quantum well exhibits quantized energy levels and a step-like density of states due to quantum confinement. This demonstrates how quantum well thickness influences the electronic properties of semiconductor devices. Simulation of density of states of quantum well structure with various nanometer thickness is accomplished.